

# POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE.

Name Krish Wadhwa

Enrollment No. NJC231090

Jaypee Institute of Information Technology, Noida

Test-1 Examination, 2023

B. Tech. I Semester

Course Title: Mathematics I

Maximum Time: 1 Hr

Course Code: 15B11MA111

Maximum Marks: 20

|     |                                                                                                                                |
|-----|--------------------------------------------------------------------------------------------------------------------------------|
| CO1 | Define the basics of matrices and calculus of functions of one or more variables.                                              |
| CO2 | Explain the concepts of calculus, matrices and Laplace transforms.                                                             |
| CO3 | Make use of concepts of matrices, calculus, differential equations and Laplace transforms in solving engineering problems.     |
| CO4 | Simplify and solve various problems of vector calculus, differential equations and Laplace transforms in engineering problems. |

Note: Attempt all questions. Use of basic scientific calculator is allowed.

- Find the symmetric matrix corresponding to the quadratic form  $q(x, y, z) = 4x^2 + 3y^2 + 2xy - 4xz + 2\sqrt{2}yz$ .  
[CO1 (Remembering), 2M]
- Show that the function  $f(x, y) = \begin{cases} \frac{y\sqrt{x}}{x^2+y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$  is not continuous at  $(0, 0)$ .  
[CO1 (Remembering), 2M]
- Find  $\frac{\partial^2 f}{\partial y \partial x}$  at  $(0, 0)$  for the function  $f(x, y) = \begin{cases} \frac{xy^2}{2x+3y} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$ .  
[CO1 (Remembering), 3M]
- Explain the Euler's theorem on homogeneous functions and hence evaluate  $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$  if  $u = \log_e \left( \frac{x^2 y^4}{x^2 + 7y^2} \right)$ .  
[CO2 (Understanding), 3M]
- If  $u = x + y - z$ ,  $v = x - y + z$  and  $w = x^2 + y^2 + z^2 - 2yz$  then find  $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ . Are  $u, v$  and  $w$  functionally related? If yes, find their relationship.  
[CO2 (Understanding), 3M]
- Using row echelon form, test whether the vectors  $X = (1, 1, -1)$ ,  $Y = (2, -3, 4)$  and  $Z = (4, -11, 14)$  are linearly dependent or not. If yes, find the value of  $a$  and  $b$  such that  $Z = aX + bY$ .  
[CO3 (Applying), 3M]
- Find the eigenvectors corresponding to two eigenvalues  $\lambda_1 = 1$  and  $\lambda_2 = 2$  of the matrix

$$A = \begin{bmatrix} 5 & -6 & -6 \\ -1 & 4 & 2 \\ 3 & -6 & -4 \end{bmatrix}$$

Identify the third eigenvalue  $\lambda_3$  of  $A$ . Is the matrix  $A$  diagonalizable? Justify your answer.

[CO3 (Applying), 4M]

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# POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

Name.....Krish Wadhwa.....

Enrollment No..22102060.....

Jaypee Institute of Information Technology, Noida  
Test-2 Examination, Odd 2023  
B. Tech. I Semester

Course Title: Mathematics 1  
Course Code: 15B11MA111

Maximum Time: 1 Hr  
Maximum Marks: 20

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|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| After pursuing the above mentioned course, the students will be able to - |                                                                                                                                |
| CO1                                                                       | Define the basics of matrices and calculus of functions of one or more variables.                                              |
| CO2                                                                       | Explain the concepts of calculus, matrices and Laplace Transforms.                                                             |
| CO3                                                                       | Make use of concepts of matrices, calculus, differential equations and Laplace transforms in solving engineering problems.     |
| CO4                                                                       | Simplify and solve various problems of vector calculus, differential equations and Laplace transforms in engineering problems. |

**Note: Attempt all questions. Use of basic scientific calculator is allowed.**

Q1. Evaluate the following integral using Beta/Gamma functions

$$\int_0^1 x \log\left(\frac{1}{x}\right) dx \quad [\text{CO3 (Applying), 2M}]$$

Q2. Draw the region of integration and utilize that to solve the following integral

$$\int_0^6 \int_{\frac{x}{3}}^2 x \sqrt{1+y^3} dy dx \quad [\text{CO3 (Applying), 3M}]$$

Q3. Using the transformation  $u = 3x + 2y, v = x - y$ , evaluate  $\int \int_R x dx dy$ , where  $R$  is the region bounded by the lines  $3x + 2y = 0, 3x + 2y = 6, x - y = 0$  and  $x - y = 1$ .

[CO2 (Understanding), 3M]

Q4. Find the second order Taylor's series approximation of the function  $f(x, y) = e^{2x+y}$  about the point  $(0, 0)$ . Also, obtain maximum absolute error in the region  $|x| < 0.1, |y| < 0.2$ .

[CO3 (Applying), 4M]

Q5. Find the absolute maxima and minima of the function  $f(x, y) = 2 + 2x + 2y - x^2 - y^2$  over the triangle in the first quadrant bounded by the lines  $x = 0, y = 0$  and  $x + y = 9$ .

[CO2 (Understanding), 4M]

Q6. Find the volume of the solid which is bounded by the surface  $x^2 + y^2 = 9$  and the planes  $y + z = 3, z = 0$ .

[CO3 (Applying), 4M]

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# POSSESSION OF MOBILES IN EXAM IS UFM PRACTICE

Name Krishna Wadhwa

Enrolment No. ....22193959....

Jaypee Institute of Information Technology, Noida  
End- Term Examination, ODD 2023  
B.Tech. I Sem

Course Title: Mathematics-I

Maximum Time: 2 Hrs.

Course Code: I5BI1MA111

Maximum Marks: 35

After pursuing this course, students will be able to:

|        |                                                                                                                                |               |
|--------|--------------------------------------------------------------------------------------------------------------------------------|---------------|
| C105.1 | define the basics of matrices and calculus of functions of one or more variables.                                              | Remembering   |
| C105.2 | explain the concepts of calculus, matrices and Laplace transforms.                                                             | Understanding |
| C105.3 | make use of the concepts of matrices, calculus, differential equations and Laplace transform in solving engineering problems.  | Applying      |
| C105.4 | simplify and solve various problems of vector calculus, differential equations and Laplace transforms in engineering problems. | Analyzing     |

Note: Attempt all questions. Use of basic scientific calculator is allowed.

- [CO1, Remembering, 3M] Find the solution of the following system of equations by Gauss elimination method:  
 $x + 3y + 6z = 12, x + 3y + 5z = 12, x + 6y + 7z = 18.$
- [CO3, Applying, 3M] Apply Dirichlet integral to find the volume of the solid bounded by the coordinate planes and the surface  $\left(\frac{x}{2}\right)^{1/2} + \left(\frac{y}{3}\right)^{1/2} + \left(\frac{z}{5}\right)^{1/2} = 1$  in the first octant.
- [CO4, Analyzing, 3M] A vector field is given by  $\vec{F} = (y - 3z)\hat{i} + (x + z)\hat{j} + (y - 3x)\hat{k}$ . Examine whether the given vector field  $\vec{F}$  is irrotational or not. If yes, find the velocity potential  $\phi$  of it.
- [CO2, Understanding, 3M] Explain the first shifting theorem of Laplace transform and hence find  $L\{e^{3t}\sin^2 t\}$ .
- [CO4, Analyzing, 3M] State the Convolution theorem and examine the inverse Laplace transform of  $\frac{1}{s^2(s-3)}$  using it.
- [CO2, Understanding, 4M] Explain the property of Jacobian for functionally related variables. Using Jacobian, determine functional dependence of the functions  $u = \frac{x-y}{x+y}$  and  $v = \frac{x+y}{x}$ . Are the functions related? If yes, find the relation.
- [CO2, Understanding, 4M] Interpret the extreme values of  $f(x, y) = x^2 + 2y^2$  on the circle  $x^2 + y^2 = 1$  by Lagrange method of multiplier.
- [CO3, Applying, 4M] Apply the concept of the line integral to evaluate  $\oint_C (xy^2 dx + x^2 dy)$ , where  $C$  is the boundary of the triangle with vertices  $P(0, 0)$ ,  $Q(1, 0)$  and  $R(1, 1)$  in counter clockwise direction.
- [CO4, Analyzing, 4M] State the Gauss Divergence theorem. Use it to simplify and evaluate  $\oiint \vec{F} \cdot \hat{n} dS$ , where  $\vec{F} = (x + z)\hat{i} + (y + z)\hat{j} + (x + y)\hat{k}$  and  $S$  is the surface of the sphere  $x^2 + y^2 + z^2 = 9$ .
- [CO3, Applying, 4M] Solve the differential equation  $\frac{d^2 y}{dx^2} + 2\frac{dy}{dx} + 10y = e^{-x}\sin 3x$ .

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